JIANGHUI SUN

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EDUCATION

University of Illinois Urbana-Champaign | Master of Computer Science (MCS) Aug 2026 - Dec 2027 (Expected)

Xi'an Jiaotong-Liverpool University | B.Sc. in Information and Computing Science Sep 2022 - Jul 2026

GPA: 3.8/4.0

RESEARCH & INDUSTRY EXPERIENCE

Research Assistant | UIUC SSAIL Present

- Investigating system optimization for prefill/decode (P/D)-disaggregated LLM inference in multi-turn agentic workloads, where repeated interactions create distinct inference and serving demands.

Research Assistant | Xi'an Jiaotong-Liverpool University Nov 2025 - Jan 2026

Fine-grained Emoji Understanding via DINOv2-Qwen-VL

- Built a multimodal emoji-understanding framework combining DINOv2 visual representations with Qwen-VL for fine-grained expression description generation.
- Designed adapter-based feature alignment to map DINOv2 representations into Qwen-VL vision feature space; connected aligned features to its language decoder to describe subtle facial expressions.

Research Assistant | Xi'an Jiaotong-Liverpool University Sep 2025 - Nov 2025

EmoSound: Emotion-Aware Audio Accompaniment

- Developed a multimodal agent for emotion-aware audio accompaniment of visual emojis for visually impaired users, integrating emotion-aligned retrieval with audio generation.
- Fine-tuned CLIP with a lightweight cross-attention soft-prompt module for low-resource emotion recognition, addressing limited emoji/emotion-aligned training data.

ML Algorithm Engineering Intern | RocKontrol Technology Group Jun 2024 - Aug 2024

- Built PyTorch forecasting models on 40K+ air-quality, emissions, and meteorological records; compared RNN, LSTM, GRU, and Transformer architectures using 24-hour sliding-window histories.
- Predicted PM2.5, O3, and AQI over 6-, 12-, and 24-hour horizons; LSTM reduced RMSE by ~12% versus baseline RNN and showed stable performance across horizons.

SELECTED RESEARCH PROJECTS

Knowledge Organization for Agent Memory Evolution | UIUC CS546 In Progress

- Studying whether semantic organization of identical knowledge improves agents ability to selectively revise memory, comparing flat and organized memory under matched content and feedback budgets.
- Designing reinforcement-learning evaluation of learning gains, stale-memory correction, and valid-knowledge preservation; planned benchmarks: STALE, AgentStream, and MemoryAgentBench.

Cross-modal Audio Generation for Visually Impaired Users Jun 2025 - Sep 2025

- Developed an image-to-audio web application using training-free multimodal RAG, a cross-modal knowledge graph, and dual-vector databases for retrieval across visual and textual information.
- Designed dual-path image processing (image-to-tags and image-to-descriptions) with Qwen-based aggregation to support emotion-aware audio generation.

PUBLICATION

Sun, J., et al. "EmoSound: A Multimodal AI Agent Framework for Emotion-Aware Audio Accompaniment of Emoticons." Advances in Brain-Inspired Cognitive Systems, LNCS 16568, Springer, 2026. Accepted; forthcoming. Presented at BICS 2025.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL

AI / ML: PyTorch, Transformers, scikit-learn, CLIP, LoRA/PEFT, multimodal learning, RAG

Tools: FastAPI, Docker, Git/GitHub, LangChain, Jupyter